

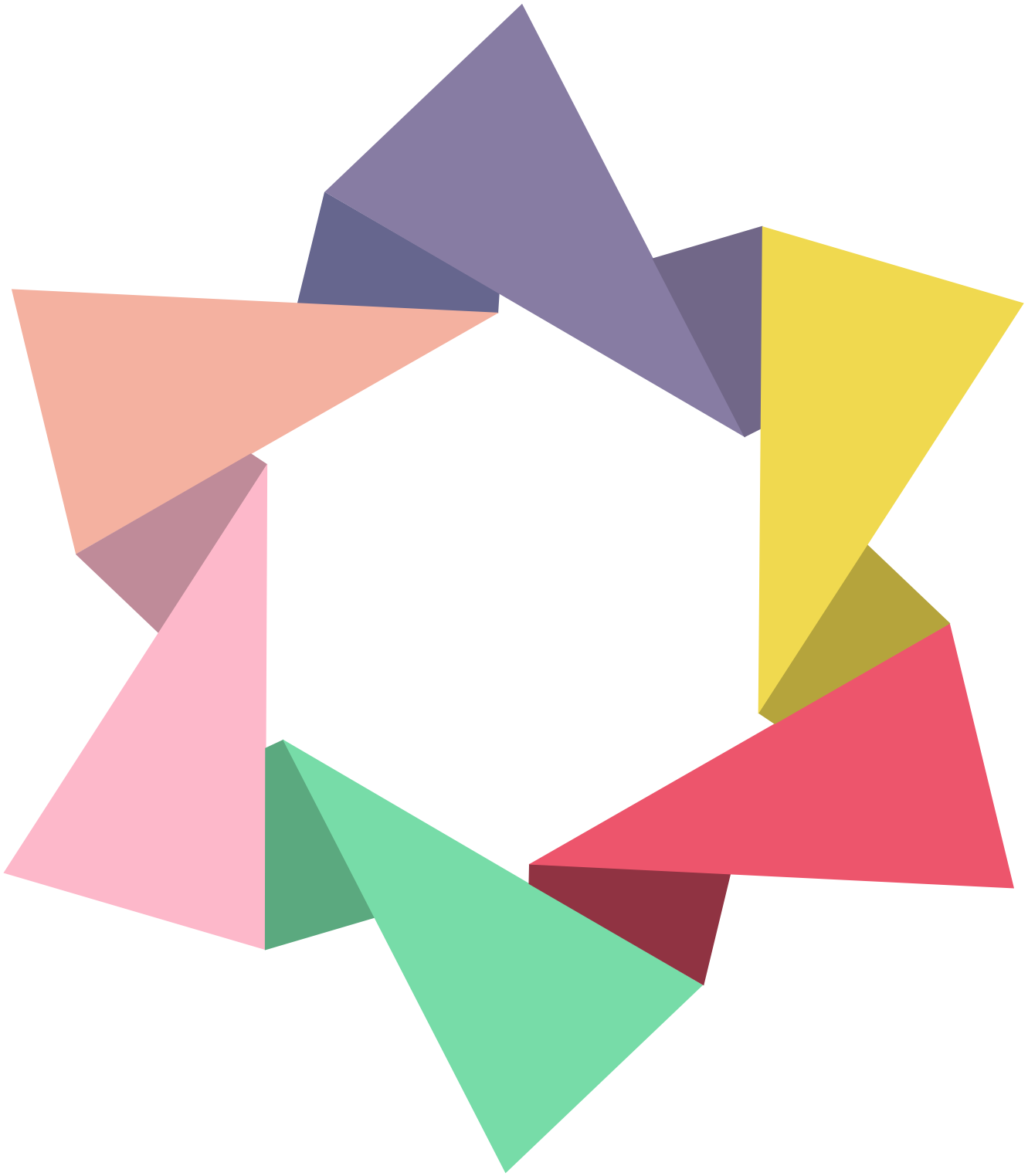
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SUPERVISED LEARNING



What Kind of Data Types?



● INFORMATION EXTRACTION

Example—Extracting information about disease outbreaks, medication dosages, or patient symptoms from electronic health records or news articles.

● NAMED ENTITY RECOGNITION (NER)

Example—Identifying names of diseases, medications, or public health organizations in a text.

● TEXT CLASSIFICATION

Example—Classifying social media posts or news articles into categories like 'disease outbreak', 'vaccine information', 'healthcare policy', etc.

● SENTIMENT ANALYSIS

Example—Analyzing public sentiment towards a new health policy or a newly released vaccine.

● DOCUMENT SUMMARIZATION

Example—Creating summaries of lengthy health reports or medical research papers.

● QUESTION ANSWERING

Example—Answering questions about a medical guideline or a health advisory.

NLP in PUBLIC HEALTH

Note—This course exclusively covers concepts and fundamental techniques in supervised learning.



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UNSUPERVISED LEARNING



Task– Discover hidden patterns or structures in the data to potentially reveal patterns like diabetes risk levels or other health conditions.

Experience– Suppose we have a dataset of individuals with the following columns:

- ▶ x_1 : Fasting blood sugar level
- ▶ x_2 : Body Mass Index (*BMI*)
- ▶ Unlike supervised learning, unsupervised learning does not use labeled outcomes like diabetes development variable.

Performance– In unsupervised learning, measuring performance is often more challenging due to the absence of a ground truth for comparison. More on this in future lectures!

Formal Presentation– We represent our dataset as a collection of pairs: $\{x^{(1)}, x^{(2)}, \dots, x^{(n)}\}$, where:

- ▶ $x^{(i)} = [x_1^{(i)}, x_2^{(i)}]$: is a vector representing the health indicators of the i^{th} individual.